$y - f(z_n) = f'(x_n)(x - x_n)$

$-f(z_n) = f'(x_n)(z_{n+1} - x_n)$

$$z_{n+1} - x_n = -\frac{f(z_n)}{f'(x_n)}$$

$$\boxed{z_{n+1} = x_n - \frac{f(z_n)}{f'(x_n)}}$$

$x_n > x_{n+1}$ (descrescător)

z_n - crescător

$|x_{n+1} - z_{n+1}| < \varepsilon$

$$\boxed{\alpha = \frac{x_{n+1} + z_{n+1}}{2}} \to \text{rădăcina}$$

C13 Metode de rezolvare a ecuațiilor diferențiale

$y' = f(x, y)$

$f : D \subset \mathbb{R}^2 \to \mathbb{R}$

$$(1)\begin{cases} y(x_0) = y^0 \\ y' = f(x, y) \end{cases}$$

(1) - Problema lui Cauchy

T (de existență și unicitate): Fie $f : D \subset \mathbb{R}^2 \to \mathbb{R}$ continuă pe D. Dacă f este lipschitziană în raport cu y, atunci $\exists$ $(x_0 - \varepsilon, x_0 + \varepsilon)$ și o funcție $y : (x_0 - \varepsilon, x_0 + \varepsilon)$ a.î. $\begin{cases} y'(x) = f(x, y(x)), & x \in (x_0 - \varepsilon, x_0 + \varepsilon) \\ y(x_0) = y^0 \end{cases}$

($\times$ lipschitziană în raport cu y:

$\forall (x, y_1), (x, y_2) \in D \; |f(x, y_1) - f(x, y_2)| \leq K |y_1 - y_2|$
$(K > 0)$)

$\left|\frac{\partial f}{\partial y}(x, y)\right| \leq K$, $K > 0$ ($|f(x, y_1) - f(x, y_2)| =$

$= |(y_2 - y_1) \frac{\partial f}{\partial y}(x, c(x))|$

$= |y_2 - y_1| \cdot \left|\frac{\partial f}{\partial y}(x, c(x))\right|$

$\leq |y_2 - y_1| K$

$y' = xy^2$, $y(0) = 1$

$f(x, y) = xy^2$

$\left|\frac{\partial f}{\partial y}\right| = |2xy| = 2|x| \cdot |y| \leq K$

$|y| \leq \frac{K}{2|x|}$

$x \in [a, b]$

$\left|\frac{\partial f}{\partial y}\right| \leq 2|b| \cdot |y| \leq K$ $\Big\} \; |y| \leq \frac{K}{2|b|}$

Metode analitice

$$y(x) = \sum_{n=0}^{\infty} a_n (x-x_0)^n, \quad a_n = \frac{y^{(n)}(x_0)}{n!}$$

$a_0 = y^0,$

$a_1 = y'(x_0) = f(x_0, y(x_0)) = f(x_0, y^0)$

$a_2 = \dots$

$y'(x) = f(x, y(x))$

$$y''(x) = \frac{\partial f}{\partial x}(x, y(x)) + \frac{\partial f}{\partial y}(x, y(x)) \cdot f(x, y(x))$$

$$y''(x_0) = \frac{\partial f}{\partial x}(x_0, y^0) + \frac{\partial f}{\partial y}(x_0, y^0) \cdot f(x_0, y^0)$$

$$y''' = \frac{\partial^2 f}{\partial x^2}(x, y(x)) + \frac{\partial^2 f}{\partial x \partial y} f(x, y(x)) +$$

$$\left[\frac{\partial^2 f}{\partial x \partial y} + \frac{\partial^2 f}{\partial y^2} f\right] f(x, y(x)) + \frac{\partial f}{\partial y^2} \dots$$

$y'' + x^2 y = e^x$

$y(0) = y'(0) = 0$

Derivăm de k ori în raport cu x

$$y^{(k+2)} + x^2 y^{(k)} + \binom{k}{1} 2x\, y^{(k-1)} + \binom{k}{2} 2 y^{(k-2)} = e^x$$

$$(f \cdot g)^{(p)} = \sum_{i=0}^{p} \binom{p}{i} f^{(p-i)} g^{(i)}, \quad g^{(0)} = g, \ f^{(0)} = f$$

Leibniz

relație de recurență

ne interesează în x = 0

$$y^{(k+2)}(0) + k(k-1)\, y^{(k-2)}(0) = 1$$

Înlocuim pe k cu 4k-2

$$y^{(4k)}(0) + (4k-2)(4k-3)\, y^{(4k-4)}(0) = 1$$

$y(0) = 0$

$y^{(4)}(0) = 1$

$\left(\left(y^{(4k)}(0) - y^{(4k-4)}(0) = 1 \right) \right)$ - ideal, exemplu de la tabla

$$y^{(4k)}(0) + (4k-2)(4k-3)\, y^{(4k-4)}(0) = 1 \;\Big|\cdot \frac{(-1)^k}{\prod_{k=1}^{n} (4k-2)(4k-3)}$$

$$a_k + \alpha_k a_{k-1} = b_k \;\Big|\cdot \frac{(-1)^k}{\alpha_1 \alpha_2 \dots \alpha_n}$$

$$\frac{(-1)^k a_k}{\alpha_1 \alpha_2 \dots \alpha_k} - \frac{(-1)^{k-1} a_{k-1}}{\alpha_1 \alpha_2 \dots \alpha_k} = \frac{(-1)^k b_k}{\alpha_1 \alpha_2 \dots \alpha_k}$$

$$C_{k+1} \quad - \quad C_k \quad = \quad \frac{(-1)^k b_k}{\alpha_1, \dots \alpha_k}$$

$$\sum_{k=0}^{n} \left(\frac{(-1)^k a_k}{\alpha_1 \dots \alpha_k} - \frac{(-1)^{k-1} a_{k-1}}{\alpha_1 \dots \alpha_k} \right) = \sum_{k=1}^{n} \frac{(-1)^k b_k}{\alpha_1 \dots \alpha_k}$$

$$\frac{(-1)^n a_n}{\alpha_1 \dots \alpha_n} - \frac{a_0}{\alpha_0} = \sum_{k=1}^{n} \frac{(-1)^k b_k}{\alpha_1 \alpha_2 \dots \alpha_k} \;\Big|\cdot \alpha_1 \alpha_2 \dots \alpha_k$$

$$a_n = \frac{(-1)^n a_0}{\alpha_0} \alpha_1 \alpha_2 \dots \alpha_n + \sum_{k=1}^{n} (-1)^{k+n} b_k \alpha_{k+1} \dots \alpha_n$$

Revenind:

$k = 4k-1$, obținem:

$$y^{(4k+1)}(0) + (4k-1)(4k-2)\, y^{(4k-3)}(0) = -1$$

Metode numerice

$$y' = f(x, y)$$
$$y(x_0) = y^0$$

$a = x_0,\ x_0 + h,\ x_0 + 2h, \dots x_0 + nh < b < x_0 + (n+1)h$

$$\left.\begin{array}{l} a + nh < b \\ a + (n+1)h > b \\ h > 0 \end{array}\right\} \left.\begin{array}{l} n \le \frac{b-a}{h} \\ n+1 \ge \frac{b-a}{h} \end{array}\right\} \Rightarrow n > \frac{b-a}{h}$$

h - pas de discretizare

$y(x_n) \to$ soluția exactă

$x_n = x_0 + nh$, $n = \dots$

$y(x_n) \to$ soluția aproximativă

$\hookrightarrow$ "y" mare

$Y(x_n) \simeq y(x_n)$; $Y_n = Y(x_n)$

$y(x_1) = Y(x_0 + h)$

Metode unipas

- Metoda lui Euler (metoda liniilor poligonale)

$$\lim_{h\to 0}\frac{y(x+h)-y(x)}{h}=y'(x)$$

$x_n,\ x_n+h$

y_n

aprox,

$$\frac{y(x_n+h)-y(x_n)}{h}\approx y'(x_n)$$

$$y_{n+1}-y_n=h\,y'(x_n)$$

$$y_{n+1}=y_n+h f(x_n,y_n)$$

$$\boxed{y_{n+1}=y_n+h f(x_n,y_n),\ n=0,1\ldots}$$

variabilă

$$y(x_n+h)=y(x_n)+y'(x_n)h+\frac{y''(x_n+\theta h)}{2!}h^2$$

$$y_{n+1}-y_n-h f(x_n,y_n)=\frac{y''(x_n+\theta h)}{2}h^2$$ de ordin h^2

$$y'=f(x,y(x))$$

$$\int_{x_{n-1}}^{x_{n+1}} y'(x)\,dx=\int_{x_{n-1}}^{x_{n+1}} f(x,y(x))\,dx$$

$$y(x_{n+1}) - y(x_{n-1}) = \underbrace{(x_{n+1} - x_{n-1}) f\left(\frac{x_n + x_{n-1}}{2}, y\left(\frac{x_n + x_{n-1}}{2}\right)\right)}$$

formulă de cuadratură

$\Rightarrow$ formula dreptunghiului

$$y_{n+1} - y_{n-1} = 2h f(x_n, y(x_n))$$

Formula punctului de mijloc

$$\boxed{y_{n+1} = y_{n-1} + 2h f(x_n, y(x_n))}$$

y_{n+1}

$x_p < x_{p+1} < \ldots < x_q$, $q < n+1$

$$y(x_q) - y(x_p) = \int_{x_p}^{x_q} f(x, y(x))\,dx = \sum_{k=p}^{q} A_k f(x_k, y_k) + R(f)$$

$q = n+1$

$$\Rightarrow y_{n+1} = y_p + \sum_{k=p}^{n+1} A_k f(x_k, y_k)$$

$$A_k = \int_{x_p}^{x_n} \frac{l(x)}{(x - x_k)\, l'(x_k)}\,dx \qquad l(x) = (x - x_p) \ldots (x - x_n)$$

$$y' = f(x, y)$$

$$y_{n+1} - y_n = \int_{x_n}^{x_{n+1}} f(x, y(x))\,dx = \frac{x_{n+1} - x_n}{2}\left[f(x_n, y(x_n)) + f(x_{n+1}, y(x_{n+1}))\right] + R$$

formula trapezului

$$y_{n+1} = y_n + \frac{h}{2}\left[f(x_n, y_n) + f(x_{n+1}, y_{n+1}) \right]$$

Metoda predictor corector

$$y_{n+1} = y_n + h f((x_n), y(x_n))$$

$$\overline{y_{n+1}} = y_n + \frac{h}{2}\left[f(x_n, y_n) + f(x_{n+1}, y_{n+1}) \right]$$